# Sensitivity analysis for contamination in egocentric-network randomized trials with interference

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**Abstract**

TEST

## 1 Introduction

- Interference in sociocentric networks is a very common setup. Difficult and expensive data acquisition. Sampled networks are often the only available information. Egocentric sampling provide a logistically feasible and accessible design to causal inference under interference.
- Missing edges due to the sampling process. leads to misspecification of the network structure. Observed exposures are not equal to true ones. Results in bias.
- Egocentric-network randomized trial (ENRT) is becoming increasingly common. This design is commonly used in various studies, such as HIV prevention (Latkin et al., 2009, 2013; Booth et al., 2016; Miller et al., 2018), substance addiction (Sherman et al., 2009), and mental health (Asmus et al., 2017; Desrosiers et al., 2020). A similar design in which the alters, and not the egos, are randomized to treatment is utilized in online A/B testing (Saint-Jacques et al., 2019).
- HPTN 037 study previously analyzed by Latkin et al. (2009, 2013). Simmons et al. (2015) assessed possible contamination between ego-networks using recall data collected from participants as validation data after the study. Their approach only considered possible contamination and did not evaluate its impact on the causal estimates.
- Egocentric-network randomized trial that assume the **non-overlapping networks** assumption (Buchanan et al., 2018, 2024; Chao et al., 2025; Fang et al., 2025). We relax it and consider similar estimands. BW: Chao et al. (2025) allow for alters with observed exposures equal one to be in fact not exposed to the treated ego (the ego didn't deliver the treatment...), but we exclude such measurement problem by our design. Namely, we only have a one-sided misclassification for alters (0 instead of 1, but not vice versa).
- Sensitivity analysis for unmeasured confounding and principal stratification under interference (VanderWeele et al., 2015). Relation to classical sensitivity analysis method (ADD REF).
- Possible extensions to Probabilistic bias analysis (PBA) – in this case, modify the title to 'Bias analysis . . . '.
- Interference with sampled data. Mostly random node or snowball sampling with linear-in-models in econometric literature.

# 2 Egocentric-network randomized trials

## 2.1 Setup and recruitment process

Let $\boldsymbol{G} = (\boldsymbol{V}, \boldsymbol{E})$ denote the population network, where $\boldsymbol{V} = \{1, \ldots, N\}$ is the set of units, and $\boldsymbol{E} = \{(i,j) : i,j \in \boldsymbol{V}\}$ is the set of edges, assumed to be binary. The population is of size $N = |\boldsymbol{V}|$, which is assumed to be a finite, unknown number. The population network $\boldsymbol{G}$ can be described by its adjacency matrix $\boldsymbol{A}$, where $A_{ij} = 1$ if edge $(i,j) \in E$, and 0 otherwise. We assume that the population network is undirected and and that there are no self-edges, that is, $A_{ij} = A_{ji}$ and $A_{ii} = 0$ for all $i$.

Units are recruited to the study via an *egocentric sampling design*. The sampling process consists of two stages. In the first stage, a set of egos (indexes) are sampled from the population. Then, in the second stage, each ego recruits a subset of its neighbors in the population network as alters (non-indexes). Formally, the recruitment process is as follows:

1. Sample egos from the population. Denote the set of recruited egos by $\mathcal{R}_e \subset \boldsymbol{V}$, and $n_e = |\mathcal{R}_e|$ as the number of egos. We do not impose any restrictions on the sampling process of egos.

2. Each ego recruits alters from its neighbors in the population network that are not recruited as egos in the first stage. We assume that each alter is recruited by a single ego, resulting in $n_e$ disjoint ego-networks. This is often enforced in the recruitment process by researchers to avoid overlaps between the observed ego-networks. Denote the set of recruited alters by $\mathcal{R}_a \subset \boldsymbol{V} \setminus \mathcal{R}_e$, and let $n_a = |\mathcal{R}_a|$ be their number.

The set of all recruited units is $\mathcal{R} = \mathcal{R}_e \cup \mathcal{R}_a$ resulting in an egocentric sample of size $n = n_e + n_a$, which is assumed to be a strict subset of the population $n \ll N$. The observed network $\widetilde{\boldsymbol{G}} = (\mathcal{R}, \widetilde{\boldsymbol{E}})$ is a subgraph of the population network $\boldsymbol{G}$ composed of $n_e$ disjoint ego-networks. The set of observed edges is $\widetilde{\boldsymbol{E}} = \cup_{i \in \mathcal{R}_e} \{(i,j) : e(j) = i\}$, where $e(j) \in \mathcal{R}_e$ is the ego of alter $j \in \mathcal{R}_a$. Clearly, $\widetilde{\boldsymbol{E}} \subseteq \boldsymbol{E}$. Let $\widetilde{\boldsymbol{A}}$ be the adjacency matrix of the observed network $\widetilde{\boldsymbol{G}}$.

Figure 1 illustrates the egocentric sampling process in a small population. Although the sampled egos are connected and some alters belong to multiple ego-networks in the population network, each ego reports only a subset of its true neighborhood, yielding an observed network made up of disjoint ego-networks.

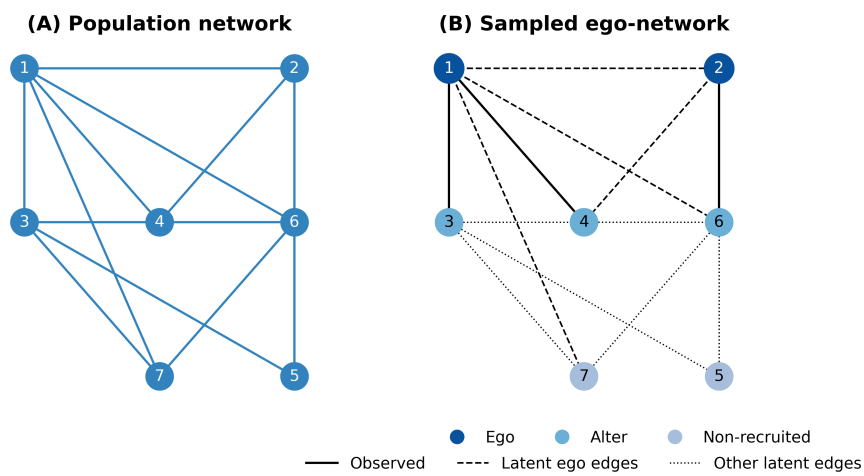


Figure 1: Egocentric sampling. Panel (A) shows the full $N = 7$ population network. Panel (B) illustrates an egocentric sample: each ego (dark blue, $n_e = 2$) reports only a subset of its true neighborhood as alters (medium blue, $n_a = 3$). The observed network is composed of disjoint ego–networks stars (solid lines), while ego–ego and unreported ego–alter connections remain latent (dashed lines). All other edges are also unobserved (dotted lines).

## 2.2 Experimental design, potential outcomes, and causal estimands

In what follows, all assumptions, estimands, and estimators are conditioned on the recruited subsample. That is, we consider in-sample causal effects. Let $\boldsymbol{Z}$ be the binary treatment assignment vector, where $Z_i = 1$ if unit $i$ is treated, and 0 otherwise. Assume that treatments are randomly assigned to the recruited egos via Bernoulli randomization.

**Assumption 1** (Bernoulli experimental design)**.** *Treatments are independently assigned only to recruited egos* $\Pr(Z_i = 1 \mid \mathcal{R}) = p_z \mathbb{I}\{i \in \mathcal{R}_e\}$ *where* $p_z \in (0,1)$ *is a known constant.*

Denote the treatment space as $\mathcal{Z} = \left\{\boldsymbol{z} \in \{0,1\}^N : \Pr(\boldsymbol{Z} = \boldsymbol{z} \mid \mathcal{R}) > 0\right\}$, where treatments of non-recruited units are assumed to be equal to zero with probability one. This experimental design, in which treatments are randomized to egos following an egocentric sample, is referred to as an *egocentric-network randomized trial* (ENRT).

Let $Y_i(\boldsymbol{z})$ be the potential outcomes of each unit under treatment $\boldsymbol{z} \in \mathcal{Z}$. We take a randomization-based inference approach, where the potential outcomes assumed to be fixed and the only source of randomness is the treatment assignment. Let $\boldsymbol{Z}_{-i}$ denote the treatment assignment of all units except $i$. Assume that $\boldsymbol{Z}_{-i}$ affects the potential outcomes of unit $i$ only through the values of the following exposure mapping (Manski, 2013; Aronow and Samii, 2017; Forastiere et al., 2020).

**Assumption 2** (Exposure mapping)**.** *Let* $F(\boldsymbol{Z}_{-i}, \boldsymbol{A}) = \mathbb{I}\left\{\sum_{j \neq i} Z_j A_{ij} > 0\right\}$*. For any* $\boldsymbol{z}, \boldsymbol{z}' \in \mathcal{Z}$*, if* $z_i = z_i'$ *and* $F(\boldsymbol{z}_{-i}, \boldsymbol{A}) = F(\boldsymbol{z}'_{-i}, \boldsymbol{A})$*, then* $Y_i(\boldsymbol{z}) = Y_i(\boldsymbol{z}')$*.*

Denote the exposures in the true population network by $F_i = F(\boldsymbol{Z}_{-i}, \boldsymbol{A})$. The exposures $F_i$ indicate whether at least one of the neighbors of unit $i$ is treated. Assumption 2 implies that the exposure mapping is correctly specified (Sävje, 2023), and that only treatments of neighbors affect the potential outcomes of unit $i$. This binary exposure mapping specification is common in the ENRT literature (Buchanan et al., 2018; Chao et al., 2025; Fang et al., 2025).

By Assumption 2, we can express the potential outcomes as $Y_i(\boldsymbol{Z}) = Y_i(Z_i, F_i)$. Consequently, each unit has four effective potential outcomes, $Y_i(z, f)$ for $z, f \in \{0,1\}$, indicating whether unit $i$ is treated and whether at least one of its neighbors is treated. We assume that the observed outcomes $Y_i,\ i \in \mathcal{R}$ are generated from one of the potential outcomes by:

**Assumption 3** (Consistency)**.** $Y_i = \sum_{z,f \in \{0,1\}} Y_i(z,f) \mathbb{I}\{Z_i = z, F_i = f\}$ *for all* $i \in \mathcal{R}$*.*

In addition, assume that we observe covariates $\boldsymbol{X}_i$ for each recruited unit.

We consider two types of causal estimands, separate for egos and alters. The first is the sample average indirect effects of treatment on the alters

$$IE = \frac{1}{n_a} \sum_{i \in \mathcal{R}_a} Y_i(0,1) - Y_i(0,0). \tag{1}$$

Eq. (1) represents the average effect on alters of being exposed to a treated neighbor compared to not being exposed. In Appendix A, we also consider the indirect effects on the relative risk scale, in the case of binary outcomes.

The second estimand is the sample average direct effect of treatment on the egos

$$DE = \frac{1}{n_e} \sum_{i \in \mathcal{R}_e} Y_i(1,0) - Y_i(0,0). \tag{2}$$

Eq. (2) represents the average effect on egos of being treated compared to not being treated when not exposed to any treated neighbors.

We explicitly separate the causal estimands for egos and alters. This separation resolve interpretation issues of other estimands (Crawford et al., 2019). Furthermore, by not conflating the two groups we avoid imposing additional assumptions and structure, as the recruitment process (Section 2.1) may induce different selection mechanisms for egos and alters, leading to incomparable distributions of potential outcomes between the two groups.

## 2.3 Estimation with the observed data

Researchers have access only to the observed data composed of the treatment assignments $\boldsymbol{Z}$, the observed outcomes $Y_i$, the observed network $\widetilde{\boldsymbol{A}}$, and possibly covariates $\boldsymbol{X}_i$. By Assumption 3, the observed outcomes $Y_i$ are generated from the potential outcomes with exposures $F_i$ computed using the *population network*. The observed exposures are $\widetilde{F}_i = F(\boldsymbol{Z}_{-i}, \widetilde{\boldsymbol{A}})$, which may differ from the true exposures $F_i$. That is, the observed network $\widetilde{\boldsymbol{A}}$ might not include all the relevant edges present in the population network $\boldsymbol{A}$ due to the egocentric sampling process, resulting in *misspecified network interference structure* (Weinstein and Nevo, 2023). We specifically care about missing ego–ego and alter–ego edges induced by the sampling process.

Consider, for example, the context of Figure 1. If only ego 1 is treated, for alter 6 we observe exposure $\widetilde{F}_i = 0$ since its ego (node 2) is not treated, but its true exposure is $F_i = 1$ due to its link to ego 1 in the population network. Similarly, for ego 2 we observe exposure $\widetilde{F}_i = 0$ while $F_i = 1$ in the population network since it is connected to ego 1.

By the ENRT design, the observed exposures of egos are $\widetilde{F}_i = 0$ for all $i \in \mathcal{R}_e$, while the observed exposures of alters can be either zero or one, depending on their ego's treatment status. Specifically, by Assumption 1, alters are exposed to a treated ego with probability $\Pr(\widetilde{F}_i = 1 \mid i \in \mathcal{R}_a) = p_z$.

Researchers can estimate causal effect only with the observed exposures. The most common estimator in randomization-based inference settings is the Horvitz-Thompson (HT) estimator. For the indirect effects (1), its form is

$$\widehat{IE} = \frac{1}{n_a} \sum_{i \in \mathcal{R}_a} \left[ \frac{\mathbb{I}\{\widetilde{F}_i = 1\} Y_i}{p_z} - \frac{\mathbb{I}\{\widetilde{F}_i = 0\} Y_i}{1 - p_z} \right], \tag{3}$$

and for direct effects (2),

$$\widehat{DE} = \frac{1}{n_e} \sum_{i \in \mathcal{R}_e} \left[ \frac{\mathbb{I}\{Z_i = 1\} Y_i}{p_z} - \frac{\mathbb{I}\{Z_i = 0\} Y_i}{1 - p_z} \right]. \tag{4}$$

If some alter–ego or ego–ego edges are missing in the observed network $\widetilde{\boldsymbol{A}}$ due to the egocentric sampling process, then for some units the observed exposures $\widetilde{F}_i$ may not agree with the true exposures $F_i$ under all possible treatment assignments $\mathcal{Z}$. In that case, both estimators (3) and (4) may not be design-unbiased.

**Proposition 1.** *Under Assumptions 1, 2, 3,*

$$\mathbb{E}_{\boldsymbol{Z}} \left[ \widehat{IE} \right] = IE + \frac{1}{n_a} \sum_{i \in \mathcal{R}_a} \frac{p_z - \pi_i^a}{1 - p_z} \left[ Y_i(0,1) - Y_i(0,0) \right],$$

$$\mathbb{E}_{\boldsymbol{Z}} \left[ \widehat{DE} \right] = DE + \frac{1}{n_e} \sum_{i \in \mathcal{R}_e} \pi_i^e \left[ \{Y_i(1,1) - Y_i(0,1)\} - \{Y_i(1,0) - Y_i(0,0)\} \right],$$

*where $\pi_i^a = \Pr(F_i = 1 \mid i \in \mathcal{R}_a)$ and $\pi_i^e = \Pr(F_i = 1 \mid i \in \mathcal{R}_e)$ are the probabilities that an alter and ego, respectively, are exposed to at least one treated neighbor in the population network $\boldsymbol{A}$.*

When an alter is connected only to its own ego, $\pi_i^a = p_z$, and (3) is a design-unbiased estimator of (1). Otherwise, whenever an alter is connected to more than one ego in the population network, $\pi_i^a > p_z$, and the indirect effect estimator (3) based on the observed exposures is biased. The magnitude of the bias depends on the number of egos each alter is connected to in the population network. That is, the more egos an alter is connected to, the larger the gap between $\pi_i^a$ and $p_z$, and the larger the bias. In addition, the naive indirect estimator (3) is attenuated towards the null of no indirect effect (Appendix A).

If an ego is not connected to any other egos in the population network, then $\pi_i^e = 0$. In that case, (4) is a design-unbiased estimator of (2). However, if an ego is connected to at least one other ego in the population network, then $\pi_i^e > 0$, and the direct effect estimator (4) based on the observed data is biased. Similarly to alters, the magnitude of the bias depend on the number ego–ego edges to in the population network.

# 3 Sensitivity analysis for contamination

Proposition 1 shows that the HT estimators based on the observed data are biased whenever there is contamination between the ego-networks. We refer to contamination as the presence alter–ego and ego–ego edges in the population network $\boldsymbol{A}$, that are missing from the observed network $\widetilde{\boldsymbol{A}}$ due to the egocentric sampling process. Under such contamination, the ego-networks are not disjoint in the population network, leading to incorrect exposures when computed using the observed network.

By the structure of the egocentric sampling and the exposure mapping specification (Assumption 2), alters with observed exposure $\widetilde{F}_i = 1$ will be correctly classified. That is $\widetilde{F}_i = 1$ implies $F_i = 1$ with probability one for $i \in \mathcal{R}_a$. However, alters with $\widetilde{F}_i = 0$ can have a true exposure of $F_i = 1$ or $F_i = 0$, depending on the structure of the population network $\boldsymbol{A}$. Therefore, the observed outcomes $Y_i$ for alters with $\widetilde{F}_i = 0$ can be generated from $Y_i(0,1)$ or $Y_i(0,0)$.

On the other hand, all egos have observed exposures of $\widetilde{F}_i = 0$, as they are linked only to alters in their ego-network in the observed network $\widetilde{\boldsymbol{A}}$. However, the observed outcomes of egos that are assigned to treatment $Z_i = z$ can be generated from $Y_i(z,1)$ or $Y_i(z,0)$, implying that observed outcomes of egos are possibly linked to four different potential outcomes, compared to only two among the alters.

We develop a sensitivity framework that enables researchers to assess the impact of possible contamination between ego-networks on the causal estimates. The framework provides bias-corrected estimators of the indirect and direct effects that account for possible exposure misspecification due to the egocentric-network recruitment process.

The bias-corrected indirect and direct effect estimators are a function of the exposure probabilities under the true population network $\pi_i^a$ and $\pi_i^e$, respectively. The structure of the egocentric sampling design and our setup allow us to decompose $\pi_i^a$ and $\pi_i^e$ into basic components that are straightforward to compute given specification of the *expected number* or the *probability* of missing alter–ego and ego–ego edges. The latter will form the basis of our sensitivity parameters, enabling seamless integration of covariates and prior knowledge into the sensitivity analysis. We provide multiple examples and practical guidelines on how to specify these parameters in Section 3.3.

For the direct effects, we require an additional sensitivity parameter to account for the complexity resulting from possibly four potential outcomes generating the observed ones. This parameter is related to the magnitude of the treatment and exposure interaction effect on the egos, and its specifics are discussed in Section 3.2.

## 3.1 Indirect effects

Define the bias-corrected indirect effect estimator as a function of the observed data and $\pi_i^a$

$$\widehat{IE}_{adj} = \frac{1}{n_a} \sum_{i \in \mathcal{R}_a} \frac{1-p_z}{1-\pi_i^a} \left[ \frac{\mathbb{I}\{\widetilde{F}_i = 1\} Y_i}{p_z} - \frac{\mathbb{I}\{\widetilde{F}_i = 0\} Y_i}{1-p_z} \right], \tag{5}$$

Given $\pi_i^a$, (5) is a design-unbiased estimator of (1)

**Proposition 2.** *Under Assumptions 1, 2, 3, we have* $\mathbb{E}_{\boldsymbol{Z}}\left[\widehat{IE}_{adj}\right] = IE$.

By the ENRT design and Assumptions 2, $\pi_i^a \geq p_z$, with equality if and only if there are no additional latent alter–ego edges. Consequently, if each alter has the same exposure probabilities $\pi_i^a = \pi^a$, then $\widehat{IE}_{adj} = \frac{1-p_z}{1-\pi^a}\widehat{IE}$. Therefore, $\text{sign}\left(\widehat{IE}_{adj}\right) = \text{sign}\left(\widehat{IE}\right)$, and $\widehat{IE}_{adj} \geq \widehat{IE}$ if $\widehat{IE} > 0$ and vice versa otherwise.

The bias-corrected estimator (5) can be modified to include an outcome model. Let $\mu_i^a(f) = \mathbb{E}\left[Y_i \mid \widetilde{F}_i = f, \boldsymbol{X}_i, i \in \mathcal{R}_a\right]$ be the conditional expectation among alters with observed exposure

$\widetilde{F}_i = f$ and covariates $\boldsymbol{X}_i$. The augmented bias-corrected estimator is

$$\widehat{IE}^{aug}_{adj} = \frac{1}{n_a} \sum_{i \in \mathcal{R}_a} \frac{1-p_z}{1-\pi^a_i} \left[ \frac{\mathbb{I}\{\widetilde{F}_i = 1\}\,(Y_i - \mu^a_i(1))}{p_z} - \frac{\mathbb{I}\{\widetilde{F}_i = 0\}\,(Y_i - \mu^a_i(0))}{1-p_z} + \{\mu^a_i(1) - \mu^a_i(0)\} \right]. \tag{6}$$

Given that $\mu^a_i(f)$ are estimated in a way that ensure their independence from the treatment assignment $\boldsymbol{Z}$, e.g., via cross-fitting (Chernozhukov et al., 2018), then (6) is also a design-unbiased estimator of (1), but can have lower variance.

Given $\pi^a_i$, the variance of the bias-corrected estimators (5) and (6) can be estimated using standard techniques for HT estimators under Bernoulli designs or thorough a nonparameteric bootstrap that resamples ego-networks (see Appendix A for details).

## 3.2 Direct effects

As the observed outcomes of egos are possibly generated from four different potential outcomes, compared to only two for the alters, bias correction for the direct effect estimator requires an additional structural assumption. Specifically, we require that the unit-level direct effects among exposed ($F_i = 1$) and non-exposed ($F_i = 0$) egos are proportional.

**Assumption 4.** *There exists a constant $\kappa$ such that $Y_i(1,1) - Y_i(0,1) = \kappa\,\{Y_i(1,0) - Y_i(0,0)\}$ for all egos $i \in \mathcal{R}_e$.*

The interpretation of the additional sensitivity parameter $\kappa$ in Assumption 4 is related to the interaction term between treatment and exposure in the potential outcome model of the egos. Consider the saturated potential outcomes model

$$Y_i(z,f) = \beta_{0i} + \beta_{1i} z + \beta_{2i} f + \beta_{3i} z f,$$

then Assumption 4 is equivalent to $\beta_{3i} = (\kappa - 1)\beta_{1i}$. That is, the interaction term $\beta_{3i}$ is proportional to the direct effect term $\beta_{1i}$ for all egos. Reasoning about the range of plausible $\kappa$ values depends on the mechanism under study. For example, in vaccine efficacy studies, if treatment represents vaccines, it is reasonable to assume that direct effects among egos with only non-vaccinated neighbors will be larger than those among egos exposed to vaccinated units, and therefore $\kappa < 1$. On the other hand, in the HPTN-037 study (Latkin et al., 2009), which we analyze in Section 5, treatment represent a behavioral intervention. Such intervention is likely to have monotonic effects, namely, the direct effect will be larger for egos exposed to a treated neighbor, leading to $\kappa > 1$.

Define the the bias-corrected direct effect estimator as a function of the observed data, $\pi^e_i$, and $\kappa$

$$\widehat{DE}_{adj} = \frac{1}{n_e} \sum_{i \in \mathcal{R}_e} \frac{1}{1 + \pi^e_i(\kappa - 1)} \left[ \frac{\mathbb{I}\{Z_i = 1\} Y_i}{p_z} - \frac{\mathbb{I}\{Z_i = 0\} Y_i}{1-p_z} \right] \tag{7}$$

Given $\pi^e_i$ and $\kappa$, (7) is a design-unbiased estimator of (2)

**Proposition 3.** *Under Assumptions 1, 2, 3, 4, we have $\mathbb{E}_{\boldsymbol{Z}}\left[\widehat{DE}_{adj}\right] = DE$.*

If $\pi^e_i = 0$ or $\kappa = 1$, the bias-corrected estimator will reduce to the naive HT estimator (4), and both will be unbiased. Namely, the naive estimator (4) is unbiased if there are no latent ego–ego edges or treatment and exposure interaction effects.

Moreover, under Assumption 4, the naive estimator $\widehat{IE}$ is attenuated *away* from the null if $\kappa > 1$, and attenuated *towards* the null if $\kappa \in \left(\min_{i \in \mathcal{R}_e}\left\{1 - \frac{1}{\pi^e_i}\right\}, 1\right)$. See Appendix A for details. However, the left bound is not informative for most plausible $\kappa$ values. For example, if $\pi^e_i = \pi^e = 0.01$, the the lower bound is $-99$, which is not practically meaningful. Thus, in general, the naive estimator is attenuated towards the null if $\kappa < 1$. In addition, for most

plausible $\kappa$ values $\text{sign}\left(\widehat{DE}_{adj}\right) = \text{sign}\left(\widehat{DE}\right)$, so the bias-corrected estimator preserves the direction of the naive estimator.

Similarly to the augmented indirect effect estimator (6), the bias-corrected direct effect estimator (7) can also be extended to include an outcome model. Let $\mu_i^e(z) = \mathbb{E}\left[Y_i \mid Z_i = z, \boldsymbol{X}_i, i \in \mathcal{R}_e\right]$. The augmented bias-corrected estimator is

$$\widehat{DE}_{adj}^{aug} = \frac{1}{n_e} \sum_{i \in \mathcal{R}_e} \frac{1}{1 + \pi_i^e(\kappa - 1)} \left[ \frac{\mathbb{I}\{Z_i = 1\}\left(Y_i - \mu_i^e(1)\right)}{p_z} - \frac{\mathbb{I}\{Z_i = 0\}\left(Y_i - \mu_i^e(0)\right)}{1 - p_z} + \{\mu_i^e(1) - \mu_i^e(0)\} \right]. \tag{8}$$

### 3.3 Specification of the sensitivity parameters

Propositions 2-3 establish that the bias-corrected estimators of the indirect and direct effects depend on the exposure probabilities $\pi_i^a = \Pr(F_i = 1 \mid i \in \mathcal{R}_a)$ and $\pi_i^e = \Pr(F_i = 1 \mid i \in \mathcal{R}_e)$ of alters and egos in the population network $\boldsymbol{A}$. For the direct effects, the bias-corrected estimator also depends on the sensitivity parameter $\kappa$. We now illustrate how the computation of $\pi_i^a$ and $\pi_i^e$ reduce to specification of edge-level probabilities or the expected number of missing alter–ego and ego–ego edges. The latter form the basis of our sensitivity parameters.

By the exposure mapping specification (Assumption 2) and the experimental design (Assumption 1), the probabilities $\pi_i^a$ and $\pi_i^e$ depend only on alter–ego or ego–ego edges in the population network $\boldsymbol{A}$. In the observed network $\widetilde{\boldsymbol{A}}$, each alter $i \in \mathcal{R}_a$ is connected only to its ego $e(i) \in \mathcal{R}_e$. Let $\mathcal{R}_e^j = \mathcal{R}_e \setminus \{j\}$ be the set of all egos other than $j \in \mathcal{R}_e$. The latent alter–ego edges $\left\{A_{ij} : i \in \mathcal{R}_a, j \in \mathcal{R}_e^{e(i)}\right\}$, and ego–ego edges $\{A_{ij} : i \neq j \in \mathcal{R}_e\}$ are the missing edges from $\widetilde{\boldsymbol{A}}$ relevant to the exposure probabilities. Define the edge-level probabilities by

$$\begin{aligned} &\Pr\left(A_{ij} = 1 \mid i \neq j \in \mathcal{R}_e\right) = \rho_{ij}^e, \\ &\Pr\left(A_{ij} = 1 \mid i \in \mathcal{R}_a, j \in \mathcal{R}_e^{e(i)}\right) = \rho_{ij}^a, \end{aligned} \tag{9}$$

for $\rho_{ij}^e, \rho_{ij}^a \in [0, 1]$. Using (9), we can derive expressions for $\pi_i^e$ and $\pi_i^a$. By the law of total probability and Assumption 1, $\Pr\left(Z_j A_{ij} = 1 \mid i \neq j \in \mathcal{R}_e\right) = p_z \rho_{ij}^e$. Assuming (conditional) edge independence yields,

$$\pi_i^e = 1 - \prod_{j \in \mathcal{R}_e^i} \left(1 - p_z \rho_{ij}^e\right). \tag{10}$$

Alter $i \in \mathcal{R}_a$ is exposed ($F_i = 1$) only if its ego $e(i)$ or at least one of the other recruited egos connected to $i$ in the population network $\boldsymbol{A}$ are treated. Therefore, we obtain (Appendix A),

$$\pi_i^a = p_z + (1 - p_z) \left[ 1 - \prod_{j \in \mathcal{R}_e^{e(i)}} \left(1 - p_z \rho_{ij}^a\right) \right]. \tag{11}$$

Eqs. (10) and (11) illustrate how the exposure probabilities can be computed given specifications of the edge-level probabilities $\rho_{ij}^e$ and $\rho_{ij}^a$. The latter form the basis of our sensitivity parameters. We now consider some prominent examples of how researchers can specify these parameters in practice. The examples include both homogeneous and heterogeneous edge probabilities, as well as specifications based on the expected number of missing edges.

**Example 1** (Homogeneous probabilities)**.** Specify the same probabilities (9) for all units: $\rho_{ij}^e = \rho^e$ and $\rho_{ij}^a = \rho^a$. The exposure probabilities (10) and (11) simplify to

$$\begin{aligned} \pi_i^e &= 1 - (1 - p_z \rho^e)^{n_e - 1}, \\ \pi_i^a &= p_z + (1 - p_z)\left[1 - (1 - p_z \rho^a)^{n_e - 1}\right]. \end{aligned}$$

For instance, if researchers suspect that 1% of the possible ego–ego edges are missing, they can set $\rho^e = 0.01$.

**Example 2** (Homogeneous number of edges)**.** A special case of Example 1 occurs by specifying the *expected number* of latent edges, separately for the ego–ego and alter–ego edges. Let $m^e \in \left\{1, \ldots, \binom{n_e}{2}\right\}$ be the expected number of ego–ego edges that are believed to be missing. Setting

$$\rho^e_{ij} = \rho^e(m^e) = \frac{m^e}{\binom{n_e}{2}},$$

weights equally each ego–ego edge probability and yield $m^e$ expected ego–ego edges.
There are $n_a(n_e - 1)$ possible missing alter–ego edges. If $m^a$ such edges are expected to be missing, set the probability of each edge to

$$\rho^a_{ij} = \rho^a(m^a) = \frac{m^a}{n_a(n_e - 1)}.$$

For instance, assume researchers suspect that each alter is connected to $k^a > 0$ additional egos on average. This can be represented by selecting $m^a = k^a \times n_a$. Similarly, if each ego is believed to be connected to $k^e > 0$ additional egos on average, then $m^e = \frac{k^e \times n_e}{2}$.

**Example 3** (Heterogeneous probabilities)**.** Examples 1-2 can be extended to incorporate unit-level characteristics through covariates $\boldsymbol{X}_i$. A well-known result in network science is homophily BW: TODO: add reference, which states that units with similar characteristics are more likely to interact. The covariates similarity can be captured through a distance function. Common examples include

- $L^p$ norm: $d(\boldsymbol{x}_i, \boldsymbol{x}_j) = \|\boldsymbol{x}_i - \boldsymbol{x}_j\|_p$.
- Cosine distance: $d(\boldsymbol{x}_i, \boldsymbol{x}_j) = 1 - \frac{\boldsymbol{x}_i^T \boldsymbol{x}_j}{\|\boldsymbol{x}_i\| \|\boldsymbol{x}_j\|}$.
- Jaccard distance: $d(\boldsymbol{x}_i, \boldsymbol{x}_j) = 1 - \frac{\|\boldsymbol{x}_i \cap \boldsymbol{x}_j\|}{\|\boldsymbol{x}_i \cup \boldsymbol{x}_j\|}$.

For $\gamma^e \geq 0$, define similarity weights using the distances $w^e_{ij} = \exp\left(-\gamma^e d(\boldsymbol{X}_i, \boldsymbol{X}_j)\right)$ or through the inner product $w^e_{ij} = \exp\left(\gamma^e \boldsymbol{X}_i^T \boldsymbol{X}_j\right)$. Then, set ego–ego edge probabilities to

$$\rho^e_{ij} = \frac{w^e_{ij}}{\sum_{j \in \mathcal{R}^i_e} w^e_{ij}}.$$

In both cases, $\gamma^e$ is the sensitivity paramaters which controls the influence of covariate similarity. Taking $\gamma^e \to 0$ yields uniform probabilities $\rho^e_{ij} \to \frac{1}{n_e - 1}$, while larger $\gamma^e$ concentrates mass on more similar nodes. Analogous definitions apply for alter–ego edges with weights $w^a_{ij}$, probabilities $\rho^a_{ij}$, and a sensitivity parameter $\gamma^a \geq 0$.

**Example 4** (Heterogeneous number of missing edges)**.** Examples 2–3 can be combined to allow pair-specific heterogeneous probabilities while fixing the expected number of missing edges. Let $w^e_{ij}$ and $w^a_{ij}$ be the weights defined in Example 3 with $\gamma^e = \gamma^a = 1$. Define the total sum of weights among ego–ego and alter–ego pairs with missing edges,

$$W^e = \sum_{i < j \in \mathcal{R}_e} w^e_{ij}, \qquad W^a = \sum_{i \in \mathcal{R}_a} \sum_{j \in \mathcal{R}^{e(i)}_e} w^a_{ij}.$$

Similarly to Example 2, specify the expected numbers $m^e$ and $m^a$ of missing ego–ego and alter–ego edges, and set

$$\rho^e_{ij} = \frac{m^e}{W^e} w^e_{ij}, \qquad \rho^a_{ij} = \frac{m^a}{W^a} w^a_{ij}.$$

This ensure that the expected number of missing edges are $m^e$ and $m^a$, while allowing for heterogeneous edge probabilities. Setting all weights ($w^e_{ij}$ and $w^a_{ij}$) to one recovers Example 2.

### 3.4 Running the sensitivity analysis in practice

To assess the the impact of contamination on the causal estimates, researchers can utilize the bias-corrected estimators (5)-(6) for the indirect effects and (7)-(8) for the direct effects. The bias-corrected estimators depend on the sensitivity parameters that define the edge-level probabilities (9) and the parameter $\kappa$ for the direct effects. Examples 1-4 provide practical guidelines on how to specify these parameters in practice. As the the true values of such parameters are unknown and can not be directly estimated from the data, researchers can resort to sensitivity analyses by exploring a range of plausible values. We consider here two approaches: grid sensitivity analysis (GSA) and probabilistic bias analysis (PBA).

In GSA, researchers specify a grid of values for the sensitivity parameters and compute the bias-corrected estimates for each combination of parameters BW: TODO: add ref. That results in a grid of bias-corrected estimates. Such estimates can be coupled with the associated confidence interval for each value of the sensitivity parameters in the specified grid.

In PBA, researchers specify a prior distribution for the sensitivity parameters and compute the distribution of bias-corrected estimates over the priors (Greenland, 2005; Lash et al., 2014; Fox, 2021). In practice, this distribution is approximated using Monte Carlo by sampling from the prior distribution, and summarized through averages and intervals of the resulting bias-corrected estimates. This distribution of bias-corrected estimates accounts for uncertainty in the sensitivity parameters. We can also compute a distribution of estimates that accounts for both design uncertainty (over the treatment assignment distribution) and bias uncertainty (from the prior of the sensitivity parameters).

We now describe the GSA and PBA approaches in more details.

**Grid sensitivity analysis.** Given a grid of sensitivity parameters, the full procedure can be performed with either analytic variance estimators or nonparametric ego-network bootstrap. In the analytic case, the steps are as follows:

(i) (Optional.) Estimate the outcome models $\mu_i^a(f)$ and $\mu_i^e(z)$. We reccomend using cross-fitting via a two-fold split. That is, estimate the outcome models on one fold and predict on the other fold, and vice versa. These outcome models are estimated only once and used for all sensitivity parameter values.

(ii) For each parameter in the grid of sensitivity parameters do:

  (a) Compute bias-corrected estimates of IE (5)-(6) and DE (7)-(8).

  (b) Estimate the variance of each bias-corrected estimator (Appendix A).

  (c) Compute confidence intervals for each estimator using normal approximation.

Alternatively, the analytic variance estimation and the normal approximation in step (ii)(b)-(c) can be replaced by a nonparametric ego-network bootstrap. That is, resample with replacement $n_e$ ego-networks from the observed data, compute the bias-corrected estimates for each bootstrap sample, and estimate the variance or confidence intervals using the empirical distribution of the bootstrap estimates. This bootstrap procedure can also be modified to account for uncertainty in the outcome models estimation by re-estimating the outcome models for each bootstrap sample.

**Probabilistic bias analysis.** Instead of specifying a grid of sensitivity parameters, researchers can specify a prior distribution for each parameter. For example, if researchers believe that on average 1% of the possible ego–ego edges are missing, they can set $\rho^e \sim \text{Beta}(a, b)$ with mean $\frac{a}{a+b} = 0.01$ in Example 1. Let $p(\cdot)$ be the prior distribution for the sensitivity parameters. In practice, researchers can specify a separate prior for each parameter.

PBA enables us to obtain a distribution of bias-corrected estimates over the prior of the sensitivity parameter. This distribution can be summarized, e.g., through average and percentiles. This distribution accounts only for the bias uncertainty (from prior of sensitivity parameters). Incorporating design uncertainty (from the distribution of $\boldsymbol{Z}$) can be achieved through random error simulation or bootstrapping (Greenland, 2005; Fox, 2021). We now describe the general structure of the PBA procedure. Let $B > 0$ be the number of Monte Carlo samples.

(i) For each $b = 1, \ldots, B$ do:

(a) Sample sensitivity parameters from the prior $p$.

(b) Compute the bias-corrected estimates $\widehat{IE}_{adj}^{(b)}$ and $\widehat{DE}_{adj}^{(b)}$ , and estimate their variance using the analytic formulas (Appendix A).

(c) Draw from the (approximate) random error distribution

$$\widehat{IE}_{adj}^{(b,rand)} \sim N\left(\widehat{IE}_{adj}^{(b)}, \widehat{\mathrm{Var}}\left(\widehat{IE}_{adj}^{(b)}\right)\right),$$

and similarly for the other estimators.

(ii) Summarize the distribution of bias-corrected estimates. For example, by computing the average $\frac{1}{B}\sum_{b=1}^{B} \widehat{IE}_{adj}^{(b,rand)}$ and empirical percentiles for intervals.

Including the outcome models can be achieved similarly to the GSA procedure by estimating them once using cross-fitting before the PBA steps. In addition, the analytic variance estimation and random error simulation in step (i)(b)-(c) can be replaced by a two-dimensional nonparametric bootstrap (Fox, 2021). That is, for each Monte Carlo sample $b = 1, \ldots, B$, resample with replacement $n_e$ ego-networks from the observed data, and compute the bias-corrected estimates on the bootstrap sample. The resulting empirical distribution of the bootstrap estimates will account for both design and bias uncertainty, and can be summarized accordingly. We provide more technical details on the motivation behind the PBA procedure in Appendix B.

# 4 Simulation study

We conduct a simulation study to evaluate the performance of the bias-corrected estimators (5)-(6) and (7)-(8). Specifically, we assess the bias, variance estimation, and empirical coverage of the confidence intervals given that the true sensitivity parameters are known. The simulation study is based on a data generating process that mimics the settings of the HPTN-037 study (Latkin et al., 2009), which we analyze in Section 5. We generated $n_e = 200$ disjoint ego-networks with two alters each, which was the average number of alters per ego-network in HPTN-037, resulting in $n_a = 400$. Contamination betweeh the ego-networks was generated by specifiying the expected number of missing alter–ego and ego–ego edges, as in Examples 2 and 4, and drawing the latent edges independently. Specifically, we set the expected number of missing alter–ego edges to $m^a \in \{100, 200, 300\}$, and ego–ego edges to $m^e \in \{50, 150, 250\}$. The binary potential outcomes were generated according to the models

$$\begin{aligned}
&\mathrm{logit}\left(\Pr\left(Y_i(z,f) = 1 \mid i \in \mathcal{R}_e\right)\right) = -2 + 2z + 0.5f + 2zf + \boldsymbol{X}_i^T \beta_e, \\
&\mathrm{logit}\left(\Pr\left(Y_i(0,f) = 1 \mid i \in \mathcal{R}_a\right)\right) = -2 + 2f + \boldsymbol{X}_i^T \beta_a,
\end{aligned}$$

implying interaction sensitivity parameter $\kappa = 2$. The data-generating proccess of the covariates $\boldsymbol{X}_i$ and the specification of $\beta_e$ and $\beta_a$ are provided in Appendix C. Treatments were assigned according to a Bernoulli design with $p_z = 0.5$. In each of the $5 \times 10^3$ simulation replications, we run the ENRT trial by generating treatment assignment. We report here the results where contamination is generated using heterogeneous edge probabilities as in Example 4 with Euclidean distance and $\gamma^e = \gamma^a = 1$. Tables **??** and **??** summarize the results for the indirect and direct effects, respectively, using both analytic and bootstrap variance estimation. We use the augmented estimators (6) and (8) with outcome models estimated by logistic regression via cross-fitting. When specification is Heterogeneous, the edge probabilities are correctly-specified, while Homogeneous uses the same probability for all edges as in Example 2. The Naive specification corresponds to the uncorrected estimators (3)-(4). As the level of contamination $(m_e, m_a)$ increase, the bias of the naive estimators increases and the coverage decreases substantially. In contrast, both bias-corrected estimators (Heterogeneous and Homogeneous) exhibit negligible bias and valid coverage across all scenarios, implying that even a misspecified model for the edge probabilities (Homogeneous) can effectively correct for the bias. Moreover, across all scenarios,

both analytic and bootstrap variance estimators over estimate the variance slightly. Additional results, including homogeneous contamination settings and non-augmented estimators, are provided in Appendix C and exhibit similar patterns.

Table 1: Simulation results for indirect effects under heterogeneous contamination with augmented estimators. Error is the empirical bias, Coverage is the empirical coverage of the 95% confidence intervals, and SD/SE is the ratio of standard deviation of the estimates to the mean standard errors. Specification "Heterogeneous" uses the correct edge probabilities, "Homogeneous" uses the same probability for all edges, and "Naive" is the uncorrected estimator.

| Scenario | Specification | Variance | Error | Coverage | SD/SE |
|---|---|---|---|---|---|
| $m_a = 100, IE = 0.358$ | Heterogeneous | Analytic | 0.016 | 0.978 | 0.774 |
| | | Bootstrap | 0.016 | 0.978 | 0.770 |
| | Homogeneous | Analytic | 0.015 | 0.979 | 0.771 |
| | | Bootstrap | 0.015 | 0.981 | 0.770 |
| | Naive | Analytic | -0.029 | 0.951 | 0.772 |
| | | Bootstrap | -0.029 | 0.951 | 0.773 |
| $m_a = 200, IE = 0.335$ | Heterogeneous | Analytic | 0.006 | 0.977 | 0.839 |
| | | Bootstrap | 0.006 | 0.977 | 0.841 |
| | Homogeneous | Analytic | 0.008 | 0.977 | 0.828 |
| | | Bootstrap | 0.008 | 0.978 | 0.829 |
| | Naive | Analytic | -0.068 | 0.678 | 0.829 |
| | | Bootstrap | -0.068 | 0.678 | 0.829 |
| $m_a = 300, IE = 0.318$ | Heterogeneous | Analytic | -0.004 | 0.972 | 0.906 |
| | | Bootstrap | -0.004 | 0.972 | 0.904 |
| | Homogeneous | Analytic | 0.003 | 0.971 | 0.902 |
| | | Bootstrap | 0.003 | 0.970 | 0.904 |
| | Naive | Analytic | -0.097 | 0.453 | 0.904 |
| | | Bootstrap | -0.098 | 0.451 | 0.902 |

Table 2: Simulation results for direct effects under heterogeneous contamination with augmented estimators. Error is the empirical bias, Coverage is the empirical coverage of the 95% confidence intervals, and SD/SE is the ratio of standard deviation of the estimates to the mean standard errors. Specification "Heterogeneous" uses the correct edge probabilities, "Homogeneous" uses the same probability for all edges, and "Naive" is the uncorrected estimator.

| Scenario | Specification | Variance | Error | Coverage | SD/SE |
|---|---|---|---|---|---|
| $m_e = 50, DE = 0.345$ | Heterogeneous | Analytic | 0.013 | 0.966 | 0.860 |
| | | Bootstrap | 0.013 | 0.964 | 0.864 |
| | Homogeneous | Analytic | 0.010 | 0.966 | 0.869 |
| | | Bootstrap | 0.010 | 0.967 | 0.864 |
| | Naive | Analytic | 0.089 | 0.684 | 0.866 |
| | | Bootstrap | 0.089 | 0.687 | 0.867 |
| $m_e = 150, DE = 0.355$ | Heterogeneous | Analytic | 0.016 | 0.936 | 0.930 |
| | | Bootstrap | 0.016 | 0.936 | 0.928 |
| | Homogeneous | Analytic | 0.015 | 0.939 | 0.930 |
| | | Bootstrap | 0.015 | 0.934 | 0.932 |
| | Naive | Analytic | 0.211 | 0.044 | 0.929 |
| | | Bootstrap | 0.210 | 0.046 | 0.935 |
| $m_e = 250, DE = 0.345$ | Heterogeneous | Analytic | 0.014 | 0.959 | 0.823 |
| | | Bootstrap | 0.014 | 0.956 | 0.832 |
| | Homogeneous | Analytic | 0.014 | 0.957 | 0.822 |
| | | Bootstrap | 0.014 | 0.956 | 0.818 |
| | Naive | Analytic | 0.270 | 0.001 | 0.820 |
| | | Bootstrap | 0.270 | 0.001 | 0.820 |

# 5 Data analysis

Analysis of HPTN 037 using the proposed SA procedures. Probably on the secondary outcomes used by others as well ("HIV risk behavior, such as sharing injection equipment"). Can take either $\rho$ to be either number of missing edges (Example 2) or with covariates (Example 3). Within the Pennsylvania site, I think they had more sub-sites, so edge probabilities can account for it, e.g., by allowing edges only between participants in the same sub-sites. Limit the analysis to a 12-month follow-up, following previous analyses (Buchanan et al., 2018, 2024; Chao et al., 2025).

## 5.1 Sensitivity analysis

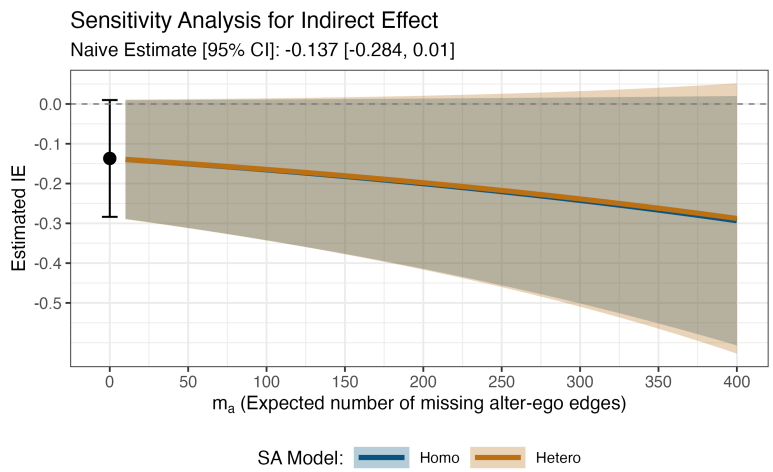


Figure 2: Caption

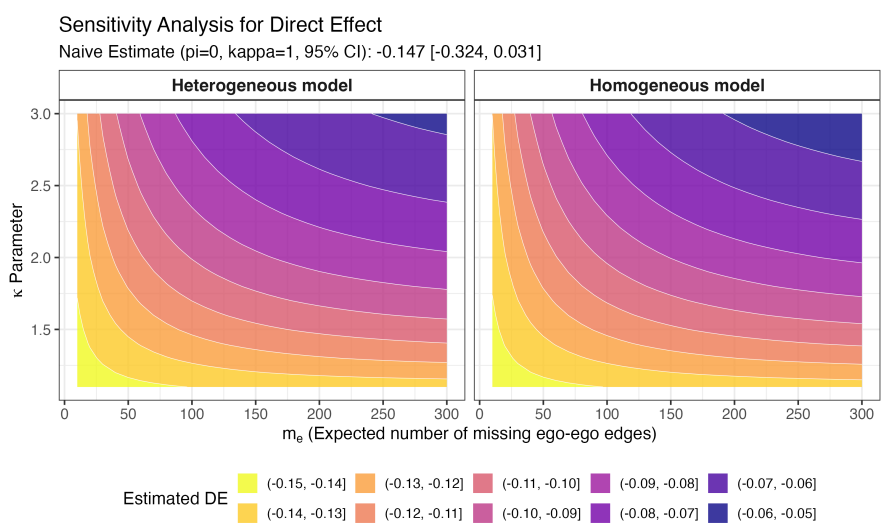


Figure 3: Caption

## 5.2 Probabilistic bias analysis

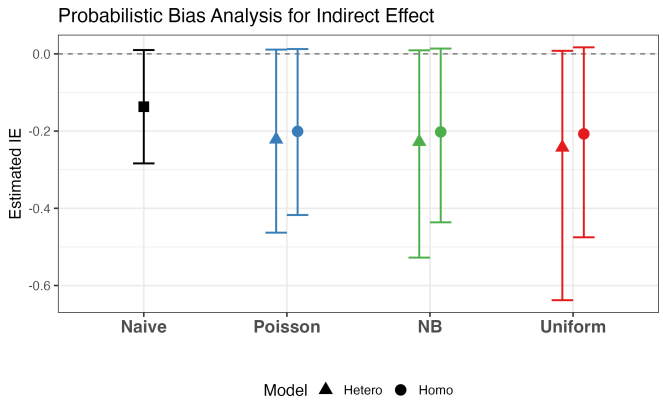


Figure 4: Caption

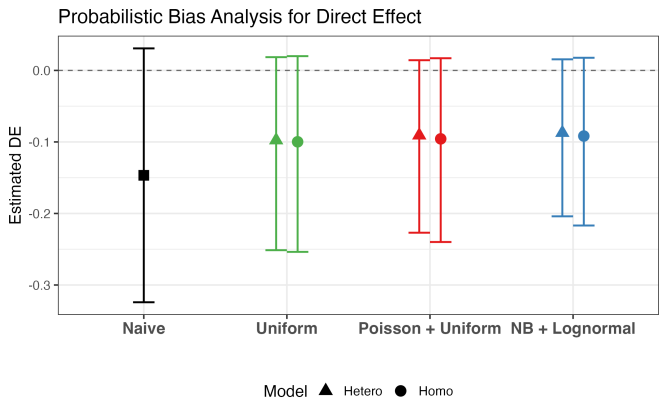


Figure 5: Caption

# 6 Discussion

1. In the case of homogeneous edge probabilities, it is possible to estimate $\pi_i^a$ (but not $\pi_i^e$) if we have validation data à la Chao et al. (2025). They did it only for IE, and only on summary-level data.

2. Extension to observational studies is tricky as non-recruited units can be treated as well... Discuss the limitations here.

3. Possible extensions to CRTs (one-stage or two-stage randomization settings). Develop the formalism and estimands for CRTs with ego-network sampling.

4. Extension to binary exposure mapping with general threshold. Can obtain similar analytical expression for probabilities of exposures.

5. Transporting the in-sample exposure-contrast estimand to population-level estimands will balance the tradeoff between the difficulty of data acquisition under interference (full population network is often required) and policy relevance.

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# A Proofs

## A.1 Indirect effects

Recall that the HT estimator of the indirect effects based on the observed data (3) is

$$\widehat{IE} = \frac{1}{n_a} \sum_{i \in \mathcal{R}_a} \left[ \frac{\mathbb{I}\{\widetilde{F}_i = 1\} Y_i}{p_z} - \frac{\mathbb{I}\{\widetilde{F}_i = 0\} Y_i}{1 - p_z} \right]$$

By the assumed experimental design (Assumption 1), the probability that an alter is assigned to treatment is zero, that is, $\Pr(Z_i = 0 \mid i \in \mathcal{R}_a) = 0$. Also, $Z_i$ and $\widetilde{F}_i = F(\boldsymbol{Z}_{-i}, \widetilde{\boldsymbol{A}})$ are independent. Thus, the event $\mathbb{I}\{\widetilde{F}_i = f\}$ is equal with probability one to $\mathbb{I}\{Z_i = 0, \widetilde{F}_i = f\}$ among the alters $i \in \mathcal{R}_a$. Starting with the first inner term, for $i \in \mathcal{R}_a$,

$$\begin{aligned}
\mathbb{E}_{\boldsymbol{Z}} \left[ \frac{\mathbb{I}\{\widetilde{F}_i = 1\} Y_i}{p_z} \right] &\overset{(i)}{=} \frac{1}{p_z} \mathbb{E}_{\boldsymbol{Z}} \left[ \mathbb{I}\{F_i = 1, \widetilde{F}_i = 1\} Y_i(0,1) + \mathbb{I}\{F_i = 0, \widetilde{F}_i = 1\} Y_i(0,0) \right] \\
&\overset{(ii)}{=} \frac{\Pr(F_i = 1, \widetilde{F}_i = 1 \mid i \in \mathcal{R}_a) Y_i(0,1)}{p_z} \\
&\quad + \frac{\Pr(F_i = 0, \widetilde{F}_i = 1 \mid i \in \mathcal{R}_a) Y_i(0,0)}{p_z} \\
&\overset{(iii)}{=} \Pr(F_i = 1 \mid \widetilde{F}_i = 1, i \in \mathcal{R}_a) Y_i(0,1) \\
&\quad + \Pr(F_i = 0 \mid \widetilde{F}_i = 1, i \in \mathcal{R}_a) Y_i(0,0) \\
&\overset{(iv)}{=} Y_i(0,1),
\end{aligned} \tag{A.1}$$

where $(i)$ follows since $\mathbb{I}\{\widetilde{F}_i = f\}$ is equal with probability one to $\mathbb{I}\{Z_i = 0, \widetilde{F}_i = f\}$ and consistency (Assumption 3) connect the observed outcomes $Y_i$ to the potential outcomes $Y_i(z, f)$; $(ii)$ from taking expectation with respect to the design $\boldsymbol{Z}$; $(iii)$ since $p_z = \Pr(\widetilde{F}_i = 1 \mid i \in \mathcal{R}_a)$ and $\frac{\Pr(F_i = f, \widetilde{F}_i = 1 \mid i \in \mathcal{R}_a)}{\Pr(\widetilde{F}_i = 1 \mid i \in \mathcal{R}_a)} = \Pr(F_i = f \mid \widetilde{F}_i = 1, i \in \mathcal{R}_a)$; and $(iv)$ since $\Pr(F_i = 1 \mid \widetilde{F}_i = 1, i \in \mathcal{R}_a) = 1$. To see this, note that $\sum_{j \neq i} \widetilde{A}_{ij} Z_j \leq \sum_{j \neq i} A_{ij} Z_j$ with probability one, which implies that $\widetilde{F}_i \leq F_i$ with probability one as well. Since $F_i$ is binary, if $\widetilde{F}_i = 1$, it must also be $F_i = 1$.

Moving to the second inner term, a similar derivation yields

$$\begin{aligned}
\mathbb{E}_{\boldsymbol{Z}} \left[ \frac{\mathbb{I}\{\widetilde{F}_i = 0\} Y_i}{1 - p_z} \right] &= \Pr(F_i = 1 \mid \widetilde{F}_i = 0, i \in \mathcal{R}_a) Y_i(0,1) \\
&\quad + \Pr(F_i = 0 \mid \widetilde{F}_i = 0, i \in \mathcal{R}_a) Y_i(0,0).
\end{aligned}$$

By Bayes' rule,

$$\begin{aligned}
\Pr(F_i = 0 \mid \widetilde{F}_i = 0, i \in \mathcal{R}_a) &= \frac{\Pr(\widetilde{F}_i = 0 \mid F_i = 0, i \in \mathcal{R}_a) \Pr(F_i = 0 \mid i \in \mathcal{R}_a)}{\Pr(\widetilde{F}_i = 0 \mid i \in \mathcal{R}_a)} \\
&= \frac{\Pr(F_i = 0 \mid i \in \mathcal{R}_a)}{\Pr(\widetilde{F}_i = 0 \mid i \in \mathcal{R}_a)} \\
&= \frac{1 - \pi_i^a}{1 - p_z},
\end{aligned}$$

as $\Pr(\widetilde{F}_i = 0 \mid F_i = 0, i \in \mathcal{R}_a) = 1$ since $\widetilde{F}_i \leq F_i$ with probability one. This yields

$$\mathbb{E}_{\boldsymbol{Z}} \left[ \frac{\mathbb{I}\{\widetilde{F}_i = 0\} Y_i}{1 - p_z} \right] = \frac{(\pi_i^a - p_z) Y_i(0,1) + (1 - \pi_i^a) Y_i(0,0)}{1 - p_z}. \tag{A.2}$$

Consequently, combining (A.1) and (A.2),

$$\begin{aligned}
\mathbb{E}_{\boldsymbol{Z}}\left[\widehat{IE}\right] &= \frac{1}{n_a}\sum_{i\in\mathcal{R}_a}\left[Y_i(0,1) - \frac{(\pi_i^a - p_z)\,Y_i(0,1) + (1-\pi_i^a)\,Y_i(0,0)}{1-p_z}\right] \\
&= \frac{1}{n_a}\sum_{i\in\mathcal{R}_a}\frac{1-\pi_i^a}{1-p_z}\left[Y_i(0,1) - Y_i(0,0)\right] \\
&= IE + \frac{1}{n_a}\sum_{i\in\mathcal{R}_a}\frac{p_z-\pi_i^a}{1-p_z}\left[Y_i(0,1) - Y_i(0,0)\right],
\end{aligned} \tag{A.3}$$

which completes the proof of the first part of Proposition 1.

The bias-corrected estimator is

$$\widehat{IE}_{adj} = \frac{1}{n_a}\sum_{i\in\mathcal{R}_a}\frac{1-p_z}{1-\pi_i^a}\left[\frac{\mathbb{I}\{\widetilde{F}_i=1\}Y_i}{p_z} - \frac{\mathbb{I}\{\widetilde{F}_i=0\}Y_i}{1-p_z}\right].$$

As the weights $\frac{1-p_z}{1-\pi_i^a}$ are fixed and therefore independent of the design $\boldsymbol{Z}$, taking the expectation $\mathbb{E}_{\boldsymbol{Z}}\left[\widehat{IE}_{adj}\right]$ will result in the second row of (A.3) where the terms $\frac{1-\pi_i^a}{1-p_z}$ and $\frac{1-p_z}{1-\pi_i^a}$ are multiplied and therefore canceled. That implies $\mathbb{E}_{\boldsymbol{Z}}\left[\widehat{IE}_{adj}\right] = IE$ and completes the proof of Proposition 2.

**Variance estimation.** We derive variance estimators based on the augmented estimator (6). Write

$$\begin{aligned}
\widehat{IE}_{adj}^{aug} &= \frac{1}{n_a}\sum_{i\in\mathcal{R}_a}\frac{1-p_z}{1-\pi_i^a}\left[\frac{\mathbb{I}\{\widetilde{F}_i=1\}\left(Y_i-\mu_i^a(1)\right)}{p_z} - \frac{\mathbb{I}\{\widetilde{F}_i=0\}\left(Y_i-\mu_i^a(0)\right)}{1-p_z} + \{\mu_i^a(1)-\mu_i^a(0)\}\right] \\
&= \frac{1}{n_a}\sum_{i\in\mathcal{R}_a}\frac{1-p_z}{1-\pi_i^a}\phi_i^a \\
&= \frac{1}{n_a}\sum_{i\in\mathcal{R}_a}\alpha_i^a\phi_i^a,
\end{aligned}$$

where $\alpha_i^a = \frac{1-p_z}{1-\pi_i^a}$. We assume that the outcome models $\mu_i^a(f)$ are estimated through cross-fitting (Chernozhukov et al., 2018) and therefore assumed to be independent of $\boldsymbol{Z}$. We ignore the sampling uncertainty associated with estimating the outcome models. Accounting for such uncertainty is possible through the nonparametric ego-network bootstrap procedure.

Alters in the same ego-network have the same observed exposure. Recall that $e(i) \in \mathcal{R}_e$ is the ego of alter $j \in \mathcal{R}_a$. We can write the adjusted IE estimator (6) as

$$\begin{aligned}
\widehat{IE}_{adj}^{aug} &= \frac{1}{n_a}\sum_{i\in\mathcal{R}_a}\alpha_i^a\phi_i^a \\
&= \frac{1}{n_a}\sum_{i\in\mathcal{R}_e}\sum_{j\in\mathcal{R}_a:e(j)=i}\alpha_j^a\phi_j^a,
\end{aligned}$$

which is the sum over all ego-networks of the sum of alters estimates within each ego-network. As estimates of alters in the same ego-network are likely correlated (often represented as the intra-cluster correlation coefficient in cluster randomized trials), we estimate the variance of $\widehat{IE}_{adj}^{aug}$ using the cluster-robust empirical variance estimator. For ego $i \in \mathcal{R}_e$, write

$$T_i = \sum_{j\in\mathcal{R}_a:e(j)=i}\alpha_j^a\phi_j^a,$$

such that

$$\widehat{IE}_{adj}^{aug} = \frac{1}{n_a}\sum_{i\in\mathcal{R}_e}T_i.$$

Let $\overline{T} = \frac{1}{n_e}\sum_{i\in\mathcal{R}_e} T_i$. Thus, $\widehat{IE}^{aug}_{adj} = \frac{n_e}{n_a}\overline{T}$. The cluster-robust empirical variance estimator is

$$\widehat{Var}\left(\widehat{IE}^{aug}_{adj}\right) = \frac{n_e}{n_a^2(n_e-1)}\sum_{i\in\mathcal{R}_e}\left(T_i - \overline{T}\right)^2.$$

Alternatively, it is also possible to estimate the variance of $\widehat{IE}^{aug}_{adj}$ by nonparametric network bootstrap, where we resample ego-networks.

**Attenuation towards the null.** Eq. (11) implies that $\pi_i^a \geq p_z$, therefore $\frac{1-p_z}{1-\pi_i^a} \geq 1$. If $\pi_i^a = \pi^a$ for all $i \in \mathcal{R}_a$, that is, the same for all alters, then $\widehat{IE}_{adj} = \frac{1-p_z}{1-\pi^a}\widehat{IE}$. Thus, $\text{sign}(\widehat{IE}_{adj}) = \text{sign}(\widehat{IE})$ and

$$\begin{cases}\widehat{IE}_{adj} \geq \widehat{IE}, & \widehat{IE} > 0,\\ \widehat{IE}_{adj} \leq \widehat{IE}, & \widehat{IE} < 0.\end{cases}$$

Moreover, if $Y_i(0,1) \geq Y_i(0,0)$ for all $i \in \mathcal{R}_a$ such that $IE \geq 0$, then since $p_z - \pi_i^a \leq 0$, from (A.3), we have $0 \leq \mathbb{E}_{\boldsymbol{Z}}\left[\widehat{IE}\right] \leq IE$. Vice versa, if $Y_i(0,1) \leq Y_i(0,0)$ for all $i \in \mathcal{R}_a$, then $IE \leq 0$ and $0 \geq \mathbb{E}_{\boldsymbol{Z}}\left[\widehat{IE}\right] \geq IE$. That is, the estimator $\widehat{IE}$ is, in expectation, attenuated towards the null of no indirect effects.

**Exposure probabilities of alters.** For alters $i \in \mathcal{R}_a$, $F_i = 1$ if its ego $e(i)$ or at least one of the other recruited egos connected to $i$ in the population network $\boldsymbol{A}$ is treated. We have

$$\begin{aligned}\pi_i^a &= \Pr(F_i = 1 \mid i \in \mathcal{R}_a)\\ &= \Pr(F_i = 1 \mid Z_{e(i)} = 1, i \in \mathcal{R}_a)\Pr(Z_{e(i)} = 1 \mid i \in \mathcal{R}_a)\\ &\quad + \Pr(F_i = 1 \mid Z_{e(i)} = 0, i \in \mathcal{R}_a)\Pr(Z_{e(i)} = 0 \mid i \in \mathcal{R}_a),\end{aligned}$$

but $\Pr(F_i = 1 \mid Z_{e(i)} = 1, i \in \mathcal{R}_a) = 1$ and $\Pr(Z_{e(i)} = 1 \mid i \in \mathcal{R}_a) = p_z\mathbb{I}\{e(i) \in \mathcal{R}_e\}$ by Assumptions 1 and 2. Since only recruited egos have a non-zero probability of being assigned to treatment, then for alter $i \in \mathcal{R}_a$, $F_i = \sum_{j\neq i} Z_j A_{ij} = \sum_{j\in\mathcal{R}_e} Z_j A_{ij}$ with probability one. Thus, $F_i = 0 \mid Z_{e(i)} = 0, i \in \mathcal{R}_a$ equal with probability one to $\mathbb{I}\{\sum_{j\in\mathcal{R}_e\setminus e(i)} Z_j A_{ij} = 0\}$. Thus,

$$\Pr(F_i = 0 \mid Z_{e(i)} = 0, i \in \mathcal{R}_a) = \prod_{j\in\mathcal{R}_e\setminus\{e(i)\}}\left(1 - p_z\rho_{ij}^a\right),$$

which yield (11).

**Extension to relative risk estimand and estimators.** In the case of binary outcomes, we can also consider the relative risk (RR) estimand

$$IE^{RR} = \frac{\sum_{i\in\mathcal{R}_a} Y_i(0,1)}{\sum_{i\in\mathcal{R}_a} Y_i(0,0)}. \tag{A.4}$$

Analogously to (3), the estimator based on the observed data is therefore

$$\widehat{IE}^{RR} = \frac{\sum_{i\in\mathcal{R}_a}\frac{\mathbb{I}\{\widetilde{F}_i=1\}Y_i}{p_z}}{\sum_{i\in\mathcal{R}_a}\frac{\mathbb{I}\{\widetilde{F}_i=0\}Y_i}{1-p_z}}. \tag{A.5}$$

We saw in (A.1) that the numerator in (A.5) is an unbiased estimator of the numerator in (A.4). However, from (A.2) the denominator is biased. Rearranging (A.2),

$$\frac{1-p_z}{1-\pi_i^a}\mathbb{E}_{\boldsymbol{Z}}\left[\frac{\mathbb{I}\{\widetilde{F}_i = 0\}Y_i}{1-p_z}\right] = Y_i(0,0) + \frac{\pi_i^a - p_z}{1-\pi_i^a}Y_i(0,1).$$

Thus, using (A.1), we can unbiasedly estimate $Y_i(0,0)$ through

$$\frac{1-p_z}{1-\pi_i^a}\frac{\mathbb{I}\{\widetilde{F}_i=0\}Y_i}{1-p_z}-\frac{\pi_i^a-p_z}{1-\pi_i^a}\frac{\mathbb{I}\{\widetilde{F}_i=1\}Y_i}{p_z}.$$

That motivates the adjusted estimator

$$\widehat{IE}_{adj}^{RR}=\frac{\sum_{i\in\mathcal{R}_a}\frac{\mathbb{I}\{\widetilde{F}_i=1\}Y_i}{p_z}}{\sum_{i\in\mathcal{R}_a}\frac{1-p_z}{1-\pi_i^a}\frac{\mathbb{I}\{\widetilde{F}_i=0\}Y_i}{1-p_z}-\frac{\pi_i^a-p_z}{1-\pi_i^a}\frac{\mathbb{I}\{\widetilde{F}_i=1\}Y_i}{p_z}}. \tag{A.6}$$

Being a ratio estimator, (A.6) is biased, but the bias can be easily bounded (Särndal et al., 2003).

## A.2 Direct effects

Recall that the HT estimator of direct effects based on the observed data (4) is

$$\widehat{DE}=\frac{1}{n_e}\sum_{i\in\mathcal{R}_e}\left[\frac{\mathbb{I}\{Z_i=1\}Y_i}{p_z}-\frac{\mathbb{I}\{Z_i=0\}Y_i}{1-p_z}\right].$$

Also, the observed exposure of all egos is zero $\Pr(\widetilde{F}_i=0\mid i\in\mathcal{R}_e)=1$. By the assumed expermential design (Assumption 1), the event $\mathbb{I}\{Z_i=z\}$ is equal with probability one to $\mathbb{I}\{Z_i=z,\widetilde{F}_i=0\}$ among the egos $i\in\mathcal{R}_e$. The expectation of the first inner term of $\widehat{DE}$, for $i\in\mathcal{R}_e$, is as follows.

$$\begin{aligned}
\mathbb{E}_{\boldsymbol{Z}}\left[\frac{\mathbb{I}\{Z_i=1\}Y_i}{p_z}\right] &\overset{(i)}{=} \mathbb{E}_{\boldsymbol{Z}}\left[\frac{\mathbb{I}\{Z_i=1,\widetilde{F}_i=0\}Y_i}{p_z}\right]\\
&\overset{(ii)}{=} \frac{1}{p_z}\mathbb{E}_{\boldsymbol{Z}}\left[\mathbb{I}\{Z_i=1,F_i=1,\widetilde{F}_i=0\}Y_i(1,1)\right]\\
&\quad+\frac{1}{p_z}\mathbb{E}_{\boldsymbol{Z}}\left[\mathbb{I}\{Z_i=1,F_i=0,\widetilde{F}_i=0\}Y_i(0,0)\right]\\
&\overset{(iii)}{=} \Pr(F_i=1,\widetilde{F}_i=0\mid i\in\mathcal{R}_e)Y_i(1,1)\\
&\quad+\Pr(F_i=0,\widetilde{F}_i=0\mid i\in\mathcal{R}_e)Y_i(1,0)\\
&\overset{(iv)}{=} \frac{\Pr(F_i=1,\widetilde{F}_i=0\mid i\in\mathcal{R}_e)}{\Pr(\widetilde{F}_i=0\mid i\in\mathcal{R}_e)}Y_i(1,1)\\
&\quad+\frac{\Pr(F_i=0,\widetilde{F}_i=0\mid i\in\mathcal{R}_e)}{\Pr(\widetilde{F}_i=0\mid i\in\mathcal{R}_e)}Y_i(1,0)\\
&=\Pr(F_i=0\mid\widetilde{F}_i=0,i\in\mathcal{R}_e)Y_i(1,1)\\
&\quad+\Pr(F_i=1\mid\widetilde{F}_i=0,i\in\mathcal{R}_e)Y_i(1,0),
\end{aligned}$$

where $(i)$ follows from the explanation described above; $(ii)$ from consistency (Assumption 3); $(iii)$ since $Z_i$ is independent from both $F_i$ and $\widetilde{F}_i$ by Assumption 1, and $\Pr(Z_i=1\mid i\in\mathcal{R}_e)=p_z$; and $(iv)$ since $\Pr(\widetilde{F}_i=0\mid i\in\mathcal{R}_e)=1$. By Bayes' rule,

$$\begin{aligned}
\Pr(F_i=1\mid\widetilde{F}_i=0,i\in\mathcal{R}_e)&=\frac{\Pr(\widetilde{F}_i=0\mid F_i=1,i\in\mathcal{R}_e)\Pr(F_i=1\mid i\in\mathcal{R}_e)}{\Pr(\widetilde{F}_i=0\mid i\in\mathcal{R}_e)}\\
&=\Pr(F_i=1\mid i\in\mathcal{R}_e),
\end{aligned}$$

since $\widetilde{F}_i=0$ with probability one for $i\in\mathcal{R}_e$, which implies that $\Pr(\widetilde{F}_i=0\mid i\in\mathcal{R}_e)$ and the conditional probability $\Pr(\widetilde{F}_i=0\mid F_i=f,i\in\mathcal{R}_e)$ equal to one. Let $\pi_i^e=\Pr(F_i=1\mid i\in\mathcal{R}_e)$. Plugging this into the above expectation yields

$$\mathbb{E}_{\boldsymbol{Z}}\left[\frac{\mathbb{I}\{Z_i=1\}Y_i}{p_z}\right]=\pi_i^eY_i(1,1)+(1-\pi_i^e)Y_i(1,0). \tag{A.7}$$

Similarly, for $z = 0$ we have

$$\mathbb{E}_{\boldsymbol{Z}}\left[\frac{\mathbb{I}\{Z_i = 0\}Y_i}{1 - p_z}\right] = \pi_i^e Y_i(0,1) + (1 - \pi_i^e)Y_i(0,0). \tag{A.8}$$

Plugging (A.7) and (A.8) into $\mathbb{E}_{\boldsymbol{Z}}\left[\widehat{DE}\right]$ yields

$$\begin{aligned}\mathbb{E}_{\boldsymbol{Z}}\left[\widehat{DE}\right] &= \frac{1}{n_e}\sum_{i\in\mathcal{R}_e}\left[\pi_i^e\left\{Y_i(1,1) - Y_i(0,1)\right\} + (1-\pi_i^e)\left\{Y_i(1,0) - Y_i(0,0)\right\}\right] \\ &= DE + \frac{1}{n_e}\sum_{i\in\mathcal{R}_e}\pi_i^e\left[\left\{Y_i(1,1) - Y_i(0,1)\right\} - \left\{Y_i(1,0) - Y_i(0,0)\right\}\right],\end{aligned} \tag{A.9}$$

which completes the second part of Proposition 1.

Now, under Assumption 4, $Y_i(1,1) - Y_i(0,1) = \kappa\left\{Y_i(1,0) - Y_i(0,0)\right\}$. Recall that the adjusted DE estimator (7) is

$$\widehat{DE}_{adj} = \frac{1}{n_e}\sum_{i\in\mathcal{R}_e}\frac{1}{1+\pi_i^e(\kappa-1)}\left[\frac{\mathbb{I}\{Z_i=1\}Y_i}{p_z} - \frac{\mathbb{I}\{Z_i=0\}Y_i}{1-p_z}\right].$$

Since both $\kappa$ and $\pi_i^e$ are fixed and therefore independent of $\boldsymbol{Z}$, we can use (A.7), (A.8), and (A.9) to obtain

$$\begin{aligned}\mathbb{E}\left[\widehat{DE}_{adj}\right] &= \frac{1}{n_e}\sum_{i\in\mathcal{R}_e}\frac{1}{1+\pi_i^e(\kappa-1)}\left[\pi_i^e\left\{Y_i(1,1) - Y_i(0,1)\right\} + (1-\pi_i^e)\left\{Y_i(1,0) - Y_i(0,0)\right\}\right] \\ &\overset{(i)}{=} \frac{1}{n_e}\sum_{i\in\mathcal{R}_e}\frac{1}{1+\pi_i^e(\kappa-1)}\left[\pi_i^e\kappa\left\{Y_i(1,0) - Y_i(0,0)\right\} + (1-\pi_i^e)\left\{Y_i(1,0) - Y_i(0,0)\right\}\right] \\ &= \frac{1}{n_e}\sum_{i\in\mathcal{R}_e}Y_i(1,0) - Y_i(0,0) \\ &= DE,\end{aligned}$$

where in $(i)$ we used Assumption 4.

**Variance estimation.** We derive variance estimators based on the augmented estimator (8). Write

$$\begin{aligned}\widehat{DE}_{adj}^{aug} &= \frac{1}{n_e}\sum_{i\in\mathcal{R}_e}\frac{1}{1+\pi_i^e(\kappa-1)}\left[\frac{\mathbb{I}\{Z_i=1\}\left(Y_i - \mu_i^e(1)\right)}{p_z} - \frac{\mathbb{I}\{Z_i=0\}\left(Y_i - \mu_i^e(0)\right)}{1-p_z} + \left\{\mu_i^e(1) - \mu_i^e(0)\right\}\right], \\ &= \frac{1}{n_a}\sum_{i\in\mathcal{R}_a}\frac{1}{1+\pi_i^e(\kappa-1)}\phi_i^e \\ &= \frac{1}{n_a}\sum_{i\in\mathcal{R}_a}\alpha_i^e\phi_i^e.\end{aligned}$$

We assume that the outcome models $\mu_i^e(z)$ are estimated through cross-fitting ( BW: TODO: add ref) and therefore assumed to be independent of $\boldsymbol{Z}$. We ignore the sampling uncertainty associated with estimating the outcome models. We estimate the variance using the empirical estimate

$$\begin{aligned}\widehat{Var}\left(\widehat{DE}_{adj}^{aug}\right) &= \frac{1}{n_e(n_e-1)}\sum_{i\in\mathcal{R}_e}\left(\alpha_i^e\phi_i^e - \frac{1}{n_a}\sum_{i\in\mathcal{R}_a}\alpha_i^e\phi_i^e\right)^2 \\ &= \frac{1}{n_e(n_e-1)}\sum_{i\in\mathcal{R}_e}\left(\alpha_i^e\phi_i^e - \widehat{DE}_{adj}^{aug}\right)^2.\end{aligned}$$

Alternatively, it is also possible to estimate the variance of $\widehat{DE}_{adj}^{aug}$ by nonparametric network bootstrap, where we resample ego-networks.

**Attenuation and interplay between $\kappa$ and $\pi_i^e$.** Consider first the case homogeneous exposure probabilities $\pi_i^e = \pi^e$ for all $i \in \mathcal{R}_e$. Then, $\widehat{DE}_{adj} = \frac{1}{1+\pi^e(\kappa-1)}\widehat{DE}$. Thus, if $1+\pi^e(\kappa-1) > 1 \Rightarrow \kappa > 1 - \frac{1}{\pi^e}$, then $\text{sign}(\widehat{DE}_{adj}) = \text{sign}(\widehat{DE})$. Otherwise, if $\kappa < 1 - \frac{1}{\pi^e}$, then $\text{sign}(\widehat{DE}_{adj}) = -\text{sign}(\widehat{DE})$. However, for example, if $\pi^e \approx 0.01$, $1 - \frac{1}{\pi^e} \approx -99$, and it is very unlikely that $\kappa$ is smaller than that value in practice. So in general, we expect that $\text{sign}(\widehat{DE}_{adj}) = \text{sign}(\widehat{DE})$.

Moreover, if $\kappa > 1$ then

$$\begin{cases} \widehat{DE}_{adj} < \widehat{DE}, & \widehat{DE} > 0, \\ \widehat{DE}_{adj} > \widehat{DE}, & \widehat{DE} < 0, \end{cases}$$

so the adjusted estimator is, in magnitude, closer to zero than the unadjusted estimator. Vice versa, if $\kappa \in \left(1 - \frac{1}{\pi^e}, 1\right)$ then

$$\begin{cases} \widehat{DE}_{adj} > \widehat{DE}, & \widehat{DE} > 0, \\ \widehat{DE}_{adj} < \widehat{DE}, & \widehat{DE} < 0, \end{cases}$$

so the adjusted estimator is, in magnitude, further away from zero than the unadjusted estimator.

Under Assumption 4, we have $Y_i(1,1) - Y_i(0,1) = \kappa\{Y_i(1,0) - Y_i(0,0)\}$. Thus, we can write (A.9) as

$$\begin{aligned} \mathbb{E}_{\boldsymbol{Z}}\left[\widehat{DE}\right] &= DE + \frac{1}{n_e}\sum_{i\in\mathcal{R}_e} \pi_i^e \left[\{Y_i(1,1) - Y_i(0,1)\} - \{Y_i(1,0) - Y_i(0,0)\}\right] \\ &= DE + \frac{1}{n_e}\sum_{i\in\mathcal{R}_e} \pi_i^e(\kappa-1)\left[Y_i(1,0) - Y_i(0,0)\right] \\ &= \frac{1}{n_e}\sum_{i\in\mathcal{R}_e} \left[1 + \pi_i^e(\kappa-1)\right]\left[Y_i(1,0) - Y_i(0,0)\right] \end{aligned}$$

Since $\pi_i^e \geq 0$, we have

- If $Y_i(1,0) \geq Y_i(0,0)$ for all $i \in \mathcal{R}_e$, then $DE \geq 0$ and

$$\begin{cases} \mathbb{E}_{\boldsymbol{Z}}\left[\widehat{DE}\right] \geq DE \geq 0, & \kappa > 1, \\ 0 \leq \mathbb{E}_{\boldsymbol{Z}}\left[\widehat{DE}\right] \leq DE, & \kappa \in \left(\min_{i\in\mathcal{R}_e}\left\{1 - \frac{1}{\pi_i^e}\right\}, 1\right), \\ \mathbb{E}_{\boldsymbol{Z}}\left[\widehat{DE}\right] \leq 0, & \kappa < \min_{i\in\mathcal{R}_e}\left\{1 - \frac{1}{\pi_i^e}\right\}. \end{cases}$$

- If $Y_i(1,0) \leq Y_i(0,0)$ for all $i \in \mathcal{R}_e$, then $DE \leq 0$ and

$$\begin{cases} \mathbb{E}_{\boldsymbol{Z}}\left[\widehat{DE}\right] \leq DE \leq 0, & \kappa > 1, \\ 0 \geq \mathbb{E}_{\boldsymbol{Z}}\left[\widehat{DE}\right] \geq DE, & \kappa \in \left(\min_{i\in\mathcal{R}_e}\left\{1 - \frac{1}{\pi_i^e}\right\}, 1\right), \\ \mathbb{E}_{\boldsymbol{Z}}\left[\widehat{DE}\right] \geq 0, & \kappa < \min_{i\in\mathcal{R}_e}\left\{1 - \frac{1}{\pi_i^e}\right\}. \end{cases}$$

Therefore, given monotonicity (in either direction) of the potential outcomes $Y_i(1,0)$ and $Y_i(0,0)$, the estimator $\widehat{DE}$ is attenuated *away* from the null of no direct effects if $\kappa > 1$ and attenuated *towards* the null if $\kappa \in \left(\min_{i\in\mathcal{R}_e}\left\{1 - \frac{1}{\pi_i^e}\right\}, 1\right)$, and changes sign if $\kappa < \min_{i\in\mathcal{R}_e}\left\{1 - \frac{1}{\pi_i^e}\right\}$. Since $\pi_i^e$ is typically small, then $\min_{i\in\mathcal{R}_e}\left\{1 - \frac{1}{\pi_i^e}\right\}$ is a large negative number, and it is unlikely that $\kappa$ is smaller than that value in practice. Thus, in practice, we expect that the estimator $\widehat{DE}$ is attenuated away from the null if $\kappa > 1$ and towards the null if $\kappa < 1$ for most realistic scenarios.

# B Probabilistic bias analysis

We give a short overview of the technical motivation behind probabilistic bias analysis (PBA). For a more detailed treatment, we refer the reader to Greenland (2005) and Fox (2021).

Assume we observe data $\boldsymbol{D}$ and are interested in estimating a causal effect $\theta$. We have an estimator $\widehat{\theta}(\boldsymbol{D})$ based on the observed data, but we know that it is biased. Namely,

$$\mathbb{E}_{\boldsymbol{D}}\left[\widehat{\theta}(\boldsymbol{D})\right] \neq \theta.$$

Assume we can find a bias parameters $\rho$ and construct an estimator $\widehat{\theta}_{adj}(\boldsymbol{D}, \rho)$ such that given $\rho$ it is unbiased (w.r.t. to the sampling distribution of the data):

$$\mathbb{E}_{\boldsymbol{D}}\left[\widehat{\theta}_{adj}(\boldsymbol{D}, \rho) \mid \rho\right] = \theta.$$

However, the bias parameters $\rho$ are typically unknown. In PBA, we specify a probability distribution for the bias parameters $\rho$ based on prior knowledge or assumptions. This distribution reflects our uncertainty about the true value of the bias parameters. Let $p(\rho)$ denote this distribution.

With a slight abuse of notation, write $\widehat{\theta}_{adj}$ as random variable in the data $\boldsymbol{D}$ and the bias parameters $\rho$ that represent the adjusted estimator given the data and bias parameters. Using this prior, we can derive a "posterior" distribution of the bias-corrected estimator $\widehat{\theta}_{adj}$ given the data $\boldsymbol{D}$ as follows.

$$p\left(\widehat{\theta}_{adj} \mid \boldsymbol{D}\right) = \int p\left(\widehat{\theta}_{adj} \mid \boldsymbol{D}, \rho\right) p(\rho) d\rho.$$

Which can be approximated through Monte Carlo integration by drawing $M$ samples $\rho^{(1)}, \ldots, \rho^{(M)}$ from $p(\rho)$ and summarizing the distribution by, e.g., the average and percentiles. To account for the sampling variability as well, note that

$$p\left(\widehat{\theta}_{adj}\right) = \int_{\boldsymbol{D}} \int_{\rho} p\left(\widehat{\theta}_{adj} \mid \boldsymbol{D}, \rho\right) p(\rho) p(\boldsymbol{D}) d\rho d\boldsymbol{D},$$

where $p(\boldsymbol{D})$ is the sampling distribution of the data. Assume we the distribution of $\widehat{\theta}_{adj}(\boldsymbol{D}, \rho^{(m)})$ with normal distribution. Then for each draw $\rho^{(m)}$ from $p(\rho)$, estimate the adjusted estimator and add random noise according to the sampling distribution to obtain a draw from $p\left(\widehat{\theta}_{adj}\right)$. Alternatively, it is possible use a two-dimensional bootstrap that, in each iteration, sample a sensitivity parameter, draw a bootstrap sample of the data, and compute the adjusted estimator in the bootstrap sample. Repeating this procedure many times allows us to approximate this integral.

In our case, the only randomness in the data is due to the randomization $\boldsymbol{Z}$. Thus, we set $\widehat{\theta}(\boldsymbol{Z}) = \widehat{IE}$ or $\widehat{DE}$, the bias parameters $\rho = \pi_i^a$ or $\pi_i^e$ and $\kappa$, and $\widehat{\theta}_{adj}(\boldsymbol{Z}, \rho) = \widehat{IE}_{adj}$ or $\widehat{DE}_{adj}$. Which motivates the PBA procedure described in Section 3.4.

# C Simulations

Table C.1: IE Results - Heterogeneous Exposure - not aug

| Scenario | Specification | Variance | Error | Coverage | SD/SE |
|---|---|---|---|---|---|
| $m_a = 100, IE = 0.358$ | Heterogeneous | Analytic | 0.016 | 0.972 | 0.855 |
| | | Bootstrap | 0.016 | 0.972 | 0.857 |
| | Homogeneous | Analytic | 0.015 | 0.973 | 0.855 |
| | | Bootstrap | 0.015 | 0.972 | 0.857 |
| | Naive | Analytic | -0.029 | 0.943 | 0.855 |
| | | Bootstrap | -0.029 | 0.943 | 0.857 |
| $m_a = 200, IE = 0.335$ | Heterogeneous | Analytic | 0.006 | 0.974 | 0.872 |
| | | Bootstrap | 0.006 | 0.972 | 0.874 |
| | Homogeneous | Analytic | 0.008 | 0.975 | 0.867 |
| | | Bootstrap | 0.008 | 0.974 | 0.869 |
| | Naive | Analytic | -0.068 | 0.831 | 0.867 |
| | | Bootstrap | -0.068 | 0.829 | 0.869 |
| $m_a = 300, IE = 0.318$ | Heterogeneous | Analytic | -0.003 | 0.975 | 0.871 |
| | | Bootstrap | -0.003 | 0.975 | 0.873 |
| | Homogeneous | Analytic | 0.003 | 0.975 | 0.872 |
| | | Bootstrap | 0.003 | 0.975 | 0.874 |
| | Naive | Analytic | -0.097 | 0.697 | 0.872 |
| | | Bootstrap | -0.097 | 0.696 | 0.874 |

Table C.2: DE Results - Heterogeneous Exposure - not aug

| Scenario | Specification | Variance | Error | Coverage | SD/SE |
|---|---|---|---|---|---|
| $m_e = 50, DE = 0.345$ | Heterogeneous | Analytic | 0.014 | 0.950 | 0.968 |
| | | Bootstrap | 0.014 | 0.950 | 0.970 |
| | Homogeneous | Analytic | 0.011 | 0.950 | 0.972 |
| | | Bootstrap | 0.011 | 0.949 | 0.975 |
| | Naive | Analytic | 0.090 | 0.776 | 0.972 |
| | | Bootstrap | 0.090 | 0.773 | 0.975 |
| $m_e = 150, DE = 0.355$ | Heterogeneous | Analytic | 0.017 | 0.932 | 1.035 |
| | | Bootstrap | 0.017 | 0.931 | 1.038 |
| | Homogeneous | Analytic | 0.016 | 0.932 | 1.042 |
| | | Bootstrap | 0.016 | 0.933 | 1.045 |
| | Naive | Analytic | 0.212 | 0.249 | 1.042 |
| | | Bootstrap | 0.212 | 0.250 | 1.045 |
| $m_e = 250, DE = 0.345$ | Heterogeneous | Analytic | 0.015 | 0.944 | 0.985 |
| | | Bootstrap | 0.015 | 0.942 | 0.988 |
| | Homogeneous | Analytic | 0.014 | 0.940 | 0.988 |
| | | Bootstrap | 0.014 | 0.940 | 0.991 |
| | Naive | Analytic | 0.271 | 0.109 | 0.988 |
| | | Bootstrap | 0.271 | 0.108 | 0.991 |

Table C.3: IE Results - homogeneous Exposure - Augmented

| Scenario | Specification | Variance | Error | Coverage | SD/SE |
|---|---|---|---|---|---|
| $m_a = 100, IE = 0.362$ | Heterogeneous | Analytic | 0.010 | 0.987 | 0.741 |
| | | Bootstrap | 0.010 | 0.988 | 0.740 |
| | Homogeneous | Analytic | 0.009 | 0.988 | 0.742 |
| | | Bootstrap | 0.009 | 0.987 | 0.745 |
| | Naive | Analytic | -0.035 | 0.943 | 0.743 |
| | | Bootstrap | -0.035 | 0.942 | 0.741 |
| $m_a = 200, IE = 0.32$ | Heterogeneous | Analytic | -0.002 | 0.976 | 0.867 |
| | | Bootstrap | -0.002 | 0.975 | 0.870 |
| | Homogeneous | Analytic | -0.001 | 0.977 | 0.865 |
| | | Bootstrap | -0.001 | 0.976 | 0.864 |
| | Naive | Analytic | -0.072 | 0.681 | 0.864 |
| | | Bootstrap | -0.071 | 0.678 | 0.864 |
| $m_a = 300, IE = 0.372$ | Heterogeneous | Analytic | -0.039 | 0.949 | 0.860 |
| | | Bootstrap | -0.039 | 0.949 | 0.861 |
| | Homogeneous | Analytic | -0.046 | 0.931 | 0.861 |
| | | Bootstrap | -0.046 | 0.931 | 0.863 |
| | Naive | Analytic | -0.148 | 0.076 | 0.864 |
| | | Bootstrap | -0.148 | 0.075 | 0.864 |

Table C.4: DE Results - homogeneous Exposure - Augmented

| Scenario | Specification | Variance | Error | Coverage | SD/SE |
|---|---|---|---|---|---|
| $m_e = 50, DE = 0.32$ | Heterogeneous | Analytic | 0.021 | 0.953 | 0.887 |
| | | Bootstrap | 0.021 | 0.954 | 0.887 |
| | Homogeneous | Analytic | 0.019 | 0.955 | 0.882 |
| | | Bootstrap | 0.019 | 0.955 | 0.885 |
| | Naive | Analytic | 0.094 | 0.666 | 0.884 |
| | | Bootstrap | 0.094 | 0.664 | 0.885 |
| $m_e = 150, DE = 0.295$ | Heterogeneous | Analytic | 0.093 | 0.297 | 0.943 |
| | | Bootstrap | 0.094 | 0.303 | 0.950 |
| | Homogeneous | Analytic | 0.088 | 0.333 | 0.957 |
| | | Bootstrap | 0.088 | 0.332 | 0.958 |
| | Naive | Analytic | 0.290 | 0.002 | 0.956 |
| | | Bootstrap | 0.290 | 0.001 | 0.957 |
| $m_e = 250, DE = 0.38$ | Heterogeneous | Analytic | -0.027 | 0.913 | 0.918 |
| | | Bootstrap | -0.027 | 0.910 | 0.920 |
| | Homogeneous | Analytic | -0.032 | 0.873 | 0.911 |
| | | Bootstrap | -0.032 | 0.877 | 0.912 |
| | Naive | Analytic | 0.216 | 0.024 | 0.909 |
| | | Bootstrap | 0.217 | 0.026 | 0.909 |

# D Data analysis

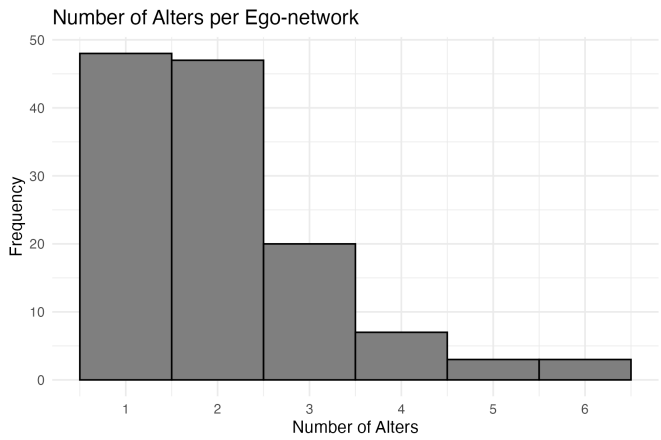


Figure D.1: Caption

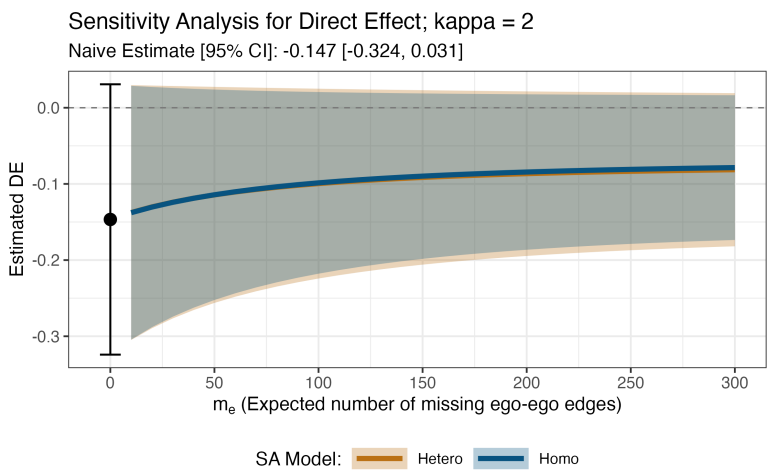


Figure D.2: Caption

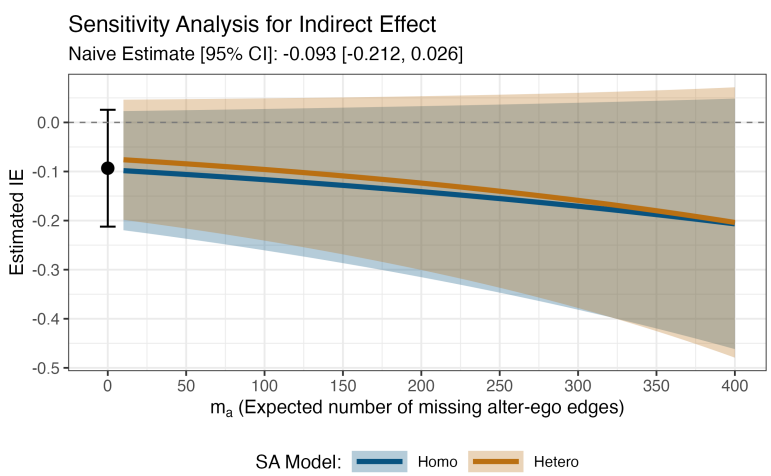


Figure D.3: SA IE Augmented

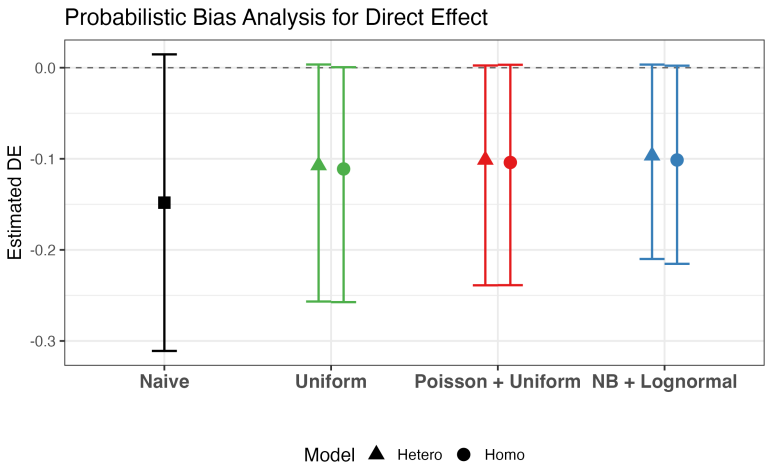


Figure D.4: PBA DE Augmented